Code No: 117CD **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, June/July - 2024
DATA WAREHOUSING AND DATA MINING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the dimension tables for library management system?	[2]
b)	State advantages and disadvantages of snowflake schema.	[3]
c)	What is data discretizaiton?	[2]
d)	What are the measures of similarity and dissimilarity?	[3]
e)	What is transactional database?	[2]
f)	What is maximal frequent itemset?	[3]
g)	Distinguish between supervised and unsupervised learning.	[2]
h)	What are the disadvantages of naïve Bayesian classifier?	[3]
i)	What are strengths and weaknesses of hierarchical clustering algorithms?	[2]
j)	Why outlier detection is important?	[3]

PART – B

(50 Marks)

2.a) Distinguish between operational database systems and data warehouse.
b) List and explain various OLAP operations. [5+5]

OR

3. Explain star schema, snowflake schema and fact constellation schema with examples. [10]

4.a) Explain the process of knowledge discovery in databases.
b) What is missing data? Explain how it is handled. [5+5]

OR

5. Explain dimensionality reduction in detail. [10]

6.a) Explain about confidence and support measures.
b) What is apriori property? Explain join and prune step in apriori algorithm. [5+5]

OR

7.a) Explain how FP growth algorithm overcomes the limitations of apriori algorithm.
b) Explain closed frequent itemset with example. [5+5]

8. Explain naïve Bayesian classifier. [10]

OR

9.a) How information gain is computed? What are the disadvantages of information gain attribute selection measure?

b) Explain Bayesian belief networks. [5+5]

10.a) Explain how clustering algorithms are evaluated.

b) What is PAM algorithm? Explain. [5+5]

OR

11. Perform k-means clustering on following data: (k=2). [10]

	x	y
A	2	3
B	1	1
C	5	6
D	3	4
E	6	7

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